Genetic Information: Cell to Gene

LAUNCH · Science · 40 minutes

I can describe where genetic information is stored and how chromosomes, DNA, genes and the genome are related.

Name: ____________________________ Date: _______________

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
chromosome	packaged DNA
gene	section of DNA
genome	all an organism's DNA

First idea

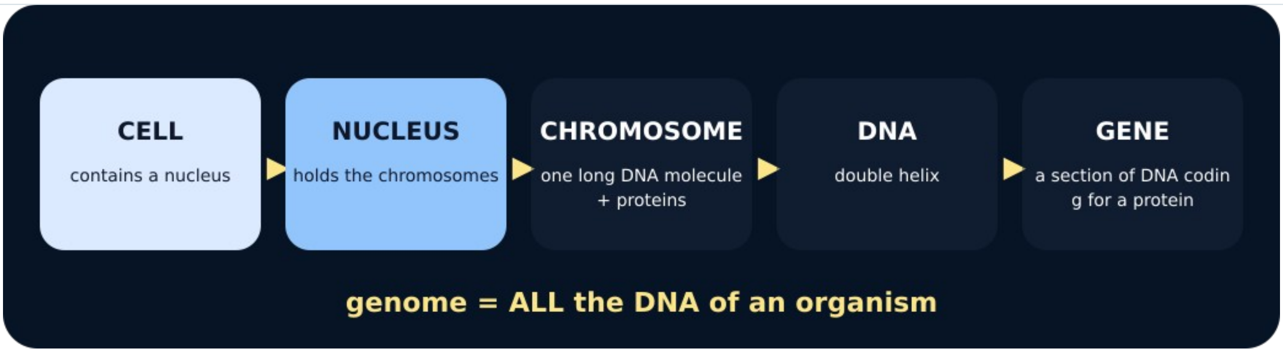
Which order moves from a large biological structure to a smaller part of its genetic material?

A. Cell → nucleus → chromosome → DNA → gene

B. Gene → cell → DNA → nucleus → chromosome

C. Cell → protein → nucleus → gene → chromosome

My first idea and reason:



Cell and nucleus

We do · choose, explain, check

Which relationship is correct?

A. A gene is a section of DNA located on a chromosome.

B. A chromosome is a section of a gene located outside the cell.

C. A genome is one protein made by one gene.

My choice: ______ Evidence or reason:

Genetic hierarchy placement lab

Place each description at the correct level, then read the complete hierarchy aloud or by pointing.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
CELL / NUCLEUS LEVEL	
CHROMOSOME / DNA LEVEL	
GENE / GENOME IDEA	

Activity cards

1 · Location	2 · Long molecule
The membrane-bound compartment that contains chromosomes in a eukaryotic cell.	The molecule associated with proteins to form a chromosome.

3 · Coding section
A section of DNA that codes for a specific protein.

Use contains and section of to describe the full hierarchy without reversing it.

Your lesson record

Build and annotate the genetic-information hierarchy, then explain one model strength and one limitation.

Model helps us see

Nested cards show part-whole relationships and let pupils physically trace from a cell to one gene.

Model does not show

The cards are not to scale, omit most molecular detail and can wrongly imply chromosomes are always X-shaped if the limitation is not stated.

Supported

Order five icon-backed cards and complete: a gene is a section of __.

Two cards are pre-positioned; use contains and part of arrows.

Standard

Build the full hierarchy and write two linked relationship sentences.

Use: The nucleus contains __. A gene is __.

Stretch

Explain the hierarchy, include genome and critique the permanent-X and scale misconceptions.

Keep to Foundation content; depth comes from relational precision and model evaluation.

Evidence target

An ordered hierarchy, accurate definitions and one honest statement about scale or shape.

Exit, Audience and evidence • W12L1

Supported: Complete: a gene is a section of __.

Standard: Describe how a chromosome, DNA and a gene are related.

Stretch: Why can the nesting model be useful even though it is not to scale?

What I said, and what it changed

____

Next I will ______.

Adult witness note

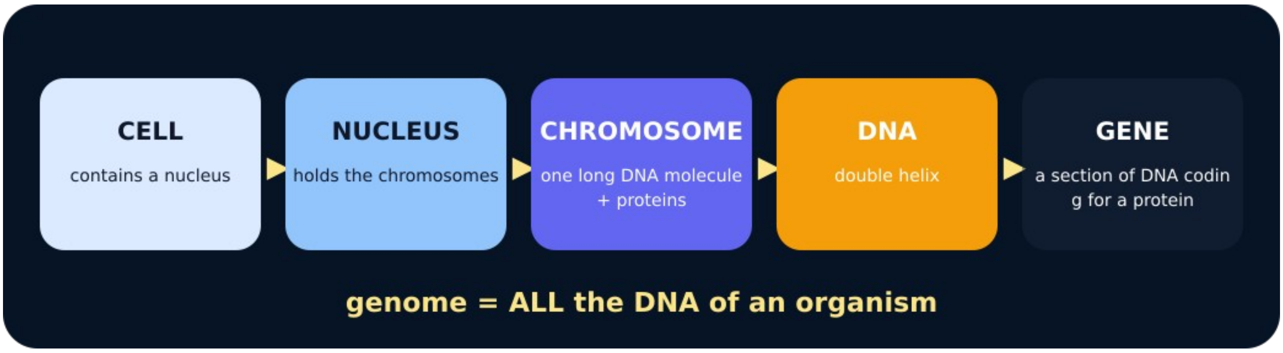
What was observed, and with what level of independence?

My evidence and explanation

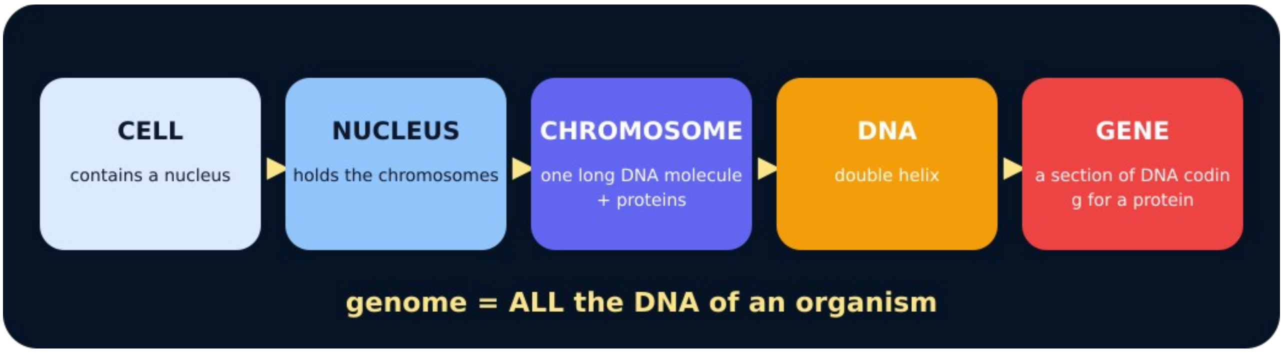
What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



To DNA



To gene

Exit • one next step

After the adult has listened, what will you keep, improve or check next?